FEDERAL SPECIFICATION

CHALK, CARPENTERS' AND RAILROAD

This specification was approved by the Director, Bureau of Federal Supply, for the use of all departments and establishments of the Government.

1. CLASSIFICATION

1.1 Types.—Chalk covered by this specification shall be of the following types, as specified:

Type I.—Carpenters' chalk.
Type II.—Railroad chalk.

1.2 Classes (for type II only).—Type II railroad chalk shall be of the following classes, as specified:

Class A.—Tapered.
Class B.—Straight sides.

1.3 Color.—The chalk shall be white or such other color as may be specified.

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

RR-S-366—Sieves; Standard, Testing.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—The chalk shall be made of such materials and by such processes as to insure compliance with the requirements of this specification, and shall be a molded homogeneous mass free from gritty particles.

3.2 Composition.

3.2.1 *White chalk.*—White chalk shall contain not less than 90 percent, by weight, calcium sulfate ($CaSO_4 2H_2O$) and not more than 5 percent free moisture as determined by the method specified in 4.3.1. The remainder shall consist of suitable binder necessary to control the degree of hardness.

3.2.2 *Colored chalk.*—Colored chalk shall contain not less than 85 percent, by weight, calcium sulfate ($CaSO_4 2H_2O$) and not more than 5 percent free moisture as determined by the method specified in 4.3.1. The remainder shall consist of colored pigment and suitable binder necessary to control the degree of hardness.

3.3 Size and shape.

3.3.1 *Type I.*—The carpenters' chalk shall be sphero-conical in shape and shall have a base diameter of $2\frac{1}{4}$ inches $\pm$ $\frac{1}{16}$ inch and shall be $1\frac{1}{8}$ inches $\pm$ $\frac{1}{16}$ inch in height.

3.3.2 *Type II.*

3.3.2.1 *Class A.*—Class A railroad chalk shall be circular in cross-section, tapered with a diameter of 1 $\pm$ $\frac{1}{32}$ inch at the large end and $\frac{7}{8}$ $\pm$ $\frac{1}{32}$ inch at the small end, and 4 inches $\pm$ $\frac{1}{16}$ inch in length. The smaller tip shall be rounded.

3.3.2.2 *Class B.*—Class B railroad chalk shall be circular in cross-section with a diameter of 1 $\pm$ $\frac{1}{32}$ inch, and 4 inches $\pm$ $\frac{1}{16}$ inch in length.

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3.4 Hardness.

3.4.1 *Type I.*—The hardness shall be of such degree as to permit the chalk to transfer freely to a string (chalk line) drawn across its surface and to leave the string, thus making a smooth, uniform, legible mark against a smooth surface such as steel, wood, concrete, or other similar surfaces in construction work.

3.4.2 *Type II.*—The hardness shall be of such degree as to permit the chalk to make a smooth, uniform, legible mark on either wood, steel, rough metal, concrete, or other construction materials not having a glossy surface.

3.5 Weight.

3.5.1 *Type I.*—The carpenters' chalk shall weigh not less than 8 pounds a gross.

3.5.2 *Type II.*—The railroad chalk shall weigh not less than 10 pounds a gross.

3.6 Breaking strength. — The chalk shall not break or crack apart when tested as required in 4.3.2 at 70° F.

3.7 Coating.—When specified, type II chalk shall be uniformly coated with a yellow enamel for cleanliness in handling.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—One box of chalk shall be taken at random from each delivery of 1,000 boxes or fraction thereof.

4.2 Inspection. — Samples shall be examined for defects and inspected for compliance with size, shape, weight, color, and such other requirements of this specification as can be determined by visual examination.

4.3 Tests.

4.3.1 *Composition.*—Ten or more pieces of the sample chalk shall be pulverized to such fineness that the material will pass through a No. 20 U. S. Standard sieve conforming to Federal Specification RR–S–366. Approximately 10 grams of the pulverized material, accurately weighed, shall be spread in a thin layer in a suitable weighed vessel. The material, thus prepared, shall be placed in an oven and dried for 2 hours ± 5 minutes at 113° ± 2° F. (45° ± 1° C.), and then cooled in an atmosphere free from moisture and weighed again. From the loss of weight of the material, the percent of free moisture shall be calculated.

4.3.2 *Breaking strength.*

4.3.2.1 *Type I.*—The breaking strength of the carpenters' chalk shall be determined by dropping the sample pieces on any hard surface from a height of 3 feet.

4.3.2.2 *Type II.*—The railroad chalk shall be supported by two parallel pins, ¼ inch in diameter, 2½ inches apart. A load of 2 pounds shall be applied perpendicular to the axis of the chalk at the center, perpendicular to the plane of the pins through a ¼-inch diameter pin parallel to the support pins, for 5 minutes.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1. *Type I.*—One-half gross of carpenters' chalk of one color shall be packaged or cushioned in such a manner as to keep it from rattling and breaking.

5.1.2 *Type II.*—One gross of railroad chalk of one type and color shall be packaged or cushioned in such a manner as to keep it from rattling and breaking.

5.2 Packing.—The chalk shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

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5.3 Marking and labeling.

5.3.1 *Packages.* — Unless otherwise specified, each package shall be plainly marked or labeled on the outside with the name, type, color, and (where applicable) the class of the commodity, the quantity contained therein, and the name of the manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, each shipping container shall be marked with the name, type, color, and (where applicable) the class of the commodity as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Intended use.

6.1.1 *Type I.*—The carpenters' chalk covered by this specification is intended primarily for use in construction work for the purpose of transferring a mark from a length of string to the work.

6.1.2 *Type II.*—The railroad chalk covered by this specification is intended primarily for the purpose of making identification marks.

6.2 Ordering data. — Purchasers should specify the type, color, and class of type II chalk required and should exercise any desired options offered herein (See 1.1, 1.2, 1.3, 3.7, 5.3, 6.3, 7.2.5, and 7.3.4.)

6.3 Colors of chalk usually available are white, red, yellow, blue, and green

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and other publications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army.

7.2.1 *Applicable specifications and other publications.*—

National Military Establishment Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN–P–108—Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–128—Packaging and Packing for Overseas Shipment — Tape; Water Resistant, Gummed.

U. S. Specification:

100–2—Standard Specification for Marking Shipments by Contractors.

(Copies of specifications required for Army purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.)

7.2.2 *Packaging for overseas shipment.*—Chalk of like type, class, and color shall be packaged in quantities of 72 pieces in snug-fitting W6s or W6c fiberboard boxes conforming to the requirements of NME Specification JAN–P–108. The chalk shall be arranged in four layers, each layer consisting of three rows of six pieces each. The layers shall be separated by full-size fiberboard separators. Suitable cushioning shall be provided to prevent movement within the boxes. The boxes

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shall be closed in conformance with the requirements of the appendix to the box specification, and in addition, all joints and seams, including those occurring in the manufacture of the boxes, shall be covered with minimum 3-inch-wide water-resistant gummed tape conforming to the requirements of NME Specification JAN–P–128, extending the tape over all corners and edges at least 3 inches.

7.2.3 *Packing for overseas shipment.*— Chalk of like type, class, and color, packaged as described in 7.2.2, shall be packed in quantities not to exceed 70 pounds gross weight in snug-fitting style, 2, 2½, or 3 nailed wood boxes, or style A or B wood-cleated plywood boxes, conforming to the requirements of NME Specifications JAN–P–106 and JAN–P–105, respectively, for type 3 load. The boxes shall be closed and strapped in conformance with the requirements of the appendix to the applicable box specification.

7.2.4 *Marking.*—In addition to any special marking required by the contract or order, each shipping container for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2.

7.2.5 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packaged, packed, and shipping containers marked for domestic or overseas shipment. (See 5.1, 5.2, 7.2.2, 7.2.3, and 7.2.4.)

7.3 **Navy.**

7.3.1 *Applicable specifications and other publications.*—

Federal Specifications:

NN–B–591—Boxes; Wood-Cleated-Fiberboard.
NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed, and Lock-Corner.
NN–B–631 — Boxes; Wood, Wirebound, for Domestic Shipment.
LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

National Military Establishment Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.
JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment — Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.
JAN–P–138—Packaging and Packing for Overseas Shipment — Boxes, Wood, Fiberboard-Lined.
JAN–P–140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

Navy Department Specification:

General Specifications for Inspection of Material.

Bureau of Supplies and Accounts Publication:

Navy Shipment Marking Handbook.

(Copies of Federal and National Military Establishment specifications required for Navy purchases, Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained without cost upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

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7.3.2 *Packing.*

7.3.2.1 *For domestic shipment.*—The chalk, packaged as described in 5.1, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery. When wood-cleated fiberboard boxes, wood-cleated plywood boxes, nailed wood boxes, wire-bound wood boxes, corrugated fiber boxes, or solid fiber boxes are used, they shall conform to the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

7.3.2.2 *For overseas shipment.*—The chalk, packaged as described in 5.1, shall be packed in cleated-plywood boxes, nailed wood boxes, wire-bound wood boxes, fiberboard boxes, or fiberboard lined wood boxes conforming to the requirements of NME Specifications JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108, and JAN–P–138, respectively. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of NME Specification JAN–P–125. The seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of NME Specification JAN–P–140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

7.3.3 *Marking.*—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with the requirements of the Navy Shipment Marking Handbook.

7.3.4 *Ordering data.* — Requests, requisitions, schedules, and contracts or orders should specify whether the subject commodity is to be packed and shipping containers marked for domestic or overseas shipment. (See 7.3.2 and 7.3.3.)

NAVY: AMCSY

7.4 **Air Force.**—Exterior shipping containers for the Air Force shall be marked in accordance with U. S. Army Specification 94–40645 Marking; Exterior, Domestic and Export Shipment, by Contractors, copies of which may be obtained from the Commanding General, Air Matériel Command, Wright-Patterson Air Force Base, Dayton, Ohio.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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